

A Perturbation Ladder for Sparse Autoencoder Dictionaries: Seed, Optimizer and Precision Against a Measured Ceiling

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DRAFT — 19 unmeasured values remain.

Every red bracket marks a number the experiment has not yet returned. No value in this document is estimated, extrapolated or illustrative: a number is either measured or visibly absent.

Abstract

This paper measures how much of a sparse autoencoder’s learned dictionary survives three perturbations—random seed, optimizer, and weight precision—using one matching procedure and one denominator, so that the three are comparable to each other.

The seed rung supplies the reference the others are read against. Two TopK autoencoders differing *only* in random seed—identical model, layer, corpus, architecture, optimizer and step count—agree on **7.9–8.3%** of their matchable features under Pearson fingerprints (8.9–10.0% under cosine), recovered at **108–112×** a permutation null. Agreement is therefore strongly significant and, simultaneously, far from total: retraining on identical data reproduces under a tenth of a dictionary. Eight percent is the ceiling no perturbation can exceed. Against it we measure optimizer (Adam versus Muon) and precision (bf16 versus locally converted 4-bit), and find **[PENDING: headline: where precision and optimizer land relative to the 8% ceiling and the 0.18–0.33% cross-architecture floor]**.

The precision rung addresses a question that arises because interpretability on consumer hardware is usually done on quantized weights. Duan [3] approach it by holding a pretrained autoencoder fixed and applying it to full-precision and quantized activations, reporting that roughly 62% (Pythia-70M) and 51% (Gemma-2-2B) of active features survive INT6 at correlation above 0.9, and identify comparison against autoencoder *training* variability as a complementary measurement their design does not make. That is the measurement reported here. The two designs answer different questions—whether a fixed dictionary’s features still fire the same way, versus whether the dictionary one would learn is the same dictionary—and their percentages are quantities over different denominators that should not be compared numerically.

Code, checkpoints and matching outputs are released; all runs are on a single Apple M1 Max.

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1 Introduction

Sparse autoencoders are trained on the residual stream of a model that a researcher could afford to run. In practice that means quantized weights. Our own prior cross-architecture study [6] had to disclose that one of its three arms was quantized while the others were not, and could not say what that cost.

1.1 Two Different Questions

There are two ways to ask whether quantization damages interpretability, and they are not the same question.

Frozen dictionary. Take one trained SAE. Apply it to activations from the full-precision model and from the quantized model. Ask whether each feature fires the same way. This is Duan [3]’s design, and it isolates the effect of weight compression from anything to do with SAE training. It answers: *if I already have a dictionary, do its features still mean what they meant?*

Relearned dictionary. Train one SAE on full-precision activations and another, identically, on quantized activations. Match the two dictionaries. This is our design. It answers: *if I do interpretability on a quantized model, do I discover the same features?*

The second is the question a practitioner faces when they train an SAE on the only model that fits their machine. It is also strictly harder to interpret, because training an SAE twice does not give the same dictionary twice even with no perturbation at all—which is exactly why it needs a baseline, and why the baseline is this paper’s central object.

1.2 Why a Survival Number Needs a Ceiling

Duan report that 62.4% of Pythia features survive INT6. Read alone, the natural reaction is that roughly a third of interpretability is lost. But neither their design nor ours has any meaning without knowing what perfect conditions produce. They supply a null in the correct direction—an FP16-versus-FP16 check returning correlation 1.0 and zero damage—which establishes that their pipeline does not manufacture drift, and under a frozen dictionary that check is the right one.

For the relearned-dictionary question no such trivial null exists, because rerunning training is itself a perturbation. The ceiling has to be measured. We measure it.

1.3 Contributions

1. **A measured ceiling for dictionary agreement.** Two autoencoders differing only in seed agree on 7.9–8.3% of matchable features at $108\text{--}112\times$ the permutation null (§2.2). Any claim that a perturbation “changes the features” has to clear this.
2. **A ladder rather than a point.** Seed, optimizer and precision measured on one instrument with one denominator, so the rungs are comparable to each other (§3).
3. **The complement Duan identify as missing:** the training-variability comparison, on a locally converted deployed quantizer rather than simulated round-to-nearest (§4).

4. **An optimizer rung nobody has measured.** Whether the choice of optimizer changes the learned decomposition is, as far as we can determine, unexamined (§3.2).
5. **Activation collection is a versioned artifact** (§4.1): released collection code is not sufficient to regenerate a released activation set, and arms collected under different collector revisions are not controlled.

2 The Instrument

2.1 Matching Procedure

Features are compared by chunk-averaged activation fingerprints over a held-out corpus. A feature pair is matched if each is the other’s nearest neighbour—reciprocal matching, which prevents a single popular feature from absorbing many partners—and if similarity exceeds a threshold τ .

τ is not chosen by hand. It is set at the 99th percentile of a permutation null in which feature identity is shuffled, so that the expected number of false matches is approximately 1% of null matches. Enrichment over that null, rather than raw match count, is the significance statistic.

Dead and low-support features are excluded before matching, and all percentages are over *matchable* features rather than dictionary size. This matters: a dictionary of 16,384 with 3,300 dead and 900 low-support features has about 12,100 matchable entries, and using the nominal size would deflate every figure by roughly a quarter.

2.2 The Ceiling

Three autoencoders were trained differing only in random seed—identical model (Llama-3.2-3B, layer 14), corpus, architecture, dictionary size, K , optimizer, step count and learning-rate schedule.

Table 1: Seed control. Three pairwise comparisons among seeds 42, 123 and 456. “% matchable” is calibrated matches over matchable features; enrichment is against the permutation null at $\tau = \text{null } p_{99}$.

fingerprint	pair	τ	% matchable	enrichment
Pearson	42 vs 123	0.9827	8.06	109.1×
Pearson	42 vs 456	0.9825	8.26	112.4×
Pearson	123 vs 456	0.9830	7.93	108.3×
cosine	42 vs 123	0.9844	9.00	121.1×
cosine	42 vs 456	0.9827	9.99	135.4×
cosine	123 vs 456	0.9846	8.93	121.6×

Two readings are both correct and both necessary. The agreement is *highly significant*—two orders of magnitude above chance, so the procedure detects shared structure when shared structure exists. And the agreement is *small*: more than nine tenths of a dictionary is not reproduced by retraining on the same data with a different seed.

Consequences. First, no perturbation can be shown to damage a dictionary by more than reseeding already does; 8% is the ceiling. Second, and less comfortably, a published SAE dictionary is largely a sample from a distribution over dictionaries, and claims about “the features of layer ℓ ” inherit that.

2.3 The Floor

The same procedure applied across architectures (Llama-3.2-3B, Pythia-1.4B, Mistral-7B) returns 0.18–0.33% of matchable features [6]. Ceiling and floor together give every rung below a scale.

3 The Ladder

Table 2: Perturbation rungs. Each arm differs from the bf16/Adam reference in exactly one respect; all share corpus, model, layer, dictionary size, K , batch, steps and seed 42.

rung	what changes	agreement	vs ceiling
seed	random seed only	7.9–8.3%	— (defines it)
optimizer	Adam $\rightarrow$ Muon	[PENDING:]	[PENDING:]
precision	bf16 $\rightarrow$ local 4-bit	[PENDING:]	[PENDING:]
floor	architecture	0.18–0.33%	

3.1 Precision

The 4-bit arm is converted *locally* from the same bf16 checkpoint with `mlx_lm.convert-q-bits 4`, not downloaded. A community upload cannot be verified as to base revision or conversion settings, and a provenance gap in the one variable under test would make the experiment unreadable. This also differs from Duan [3], who simulate round-to-nearest quantization and list the absence of a deployed quantizer as a limitation.

A sanity gate aborts the run if the bf16 and 4-bit activation files hash identically, since that would mean the conversion had no effect and the comparison was vacuous.

3.2 Optimizer

Adam versus Muon [5], the orthogonalised-momentum optimizer whose scaled results at Moonshot brought it to attention.

The arms are matched on reconstruction quality, not on learning rate. Adam and Muon have natural learning rates two orders of magnitude apart, and Muon rescales internally by $\sqrt{\text{rows}/\text{cols}}$. Holding the learning rate fixed would pin a nuisance parameter and let quality float, so a low agreement score could mean nothing more than that Muon was undertrained. A short probe selects the Muon learning rate that matches Adam’s FVE under an identical schedule, and the full run uses it. This is also the control our own prior finding argues for: if standard quality metrics cannot distinguish dictionaries that disagree about most of their content [6], then equal-quality dictionaries are precisely what should be compared.

Two implementation requirements keep the arms separated by one variable. Muon’s Newton–Schulz step is defined for matrices, and MLX degrades to plain momentum SGD for 0D/1D parameters, so biases are routed to AdamW as that implementation’s documentation specifies. Muon’s weight decay is set to zero rather than its default 0.01, matching the Adam arm, which applies none.

4 Method

[PENDING: full method table: model, layer, corpus, token budget, dictionary size, K, steps, batch, schedule, hardware, wall-clock per arm]

4.1 Activation Provenance

Every arm is trained on activations collected as part of this work, from a corpus regenerated and committed here, rather than on any previously collected set.

This is a requirement, not a preference. Activations previously collected for the autoencoders of Melton [6] cannot be regenerated from the released collection script: the metadata schemas differ in field names, indicating the collector was subsequently rewritten, and a hash over the leading 3.072×10^9 bytes differs between the original and a fresh collection. An activation set that cannot be regenerated cannot serve as a controlled arm, because the one variable under test would share a file with an uncontrolled one.

The general form of the result is that **activation collection is a versioned artifact, not a script**. Releasing the code that produced a set of activations is not sufficient to reproduce them, and a pipeline that treats collection as reproducible by construction will silently compare arms that were not collected the same way.

5 Results

Table 3: Agreement by rung. [PENDING: caption to state fingerprint metric and τ]

comparison	τ	matches	% matchable	enrichment
bf16 vs 4-bit	[PENDING:]	[PENDING:]	[PENDING:]	[PENDING:]
Adam vs Muon	[PENDING:]	[PENDING:]	[PENDING:]	[PENDING:]

5.1 Dictionary Quality Across Arms

[PENDING: loss, FVE, L0, dead-feature count per arm. Prediction to be tested, not assumed: quality metrics will be near-identical across arms regardless of how far the dictionaries diverge, extending the metrics-blindness result of melton2026sae.]

6 Analysis

To be written once Table 3 is filled, and written once. The measurement selects the argument; it is not a menu to hedge across.

If precision agreement approaches the 8% ceiling: quantization perturbs the learned dictionary no more than reseeding does, and interpretability on 4-bit weights is sound to the same degree that interpretability on any single SAE is sound—which is a real qualification, but not one specific to quantization.

If it falls well below: features discovered on quantized models are substantially not the features of the full-precision model, and a large amount of consumer-hardware interpretability needs re-examination.

If the optimizer rung moves as much as the precision rung: the more uncomfortable outcome, since optimizer choice is rarely reported as a threat to validity at all.

7 Related Work

Duan [3] is the closest work and the reference point throughout; §1.1 states the difference in question, and we do not present competing numbers for the same quantity. Bricken et al. [1] and Cunningham et al. [2] establish the dictionary-learning setting; Gao et al. [4] the TopK variant used here. **[PENDING: additional coverage: SAE reproducibility and feature-stability literature under vocabulary other than “feature universality”, given that phrase-count surveys proved unreliable for this purpose in companion work].**

8 Limitations

One model, one layer. Llama-3.2-3B layer 14. Whether the ceiling is 8% elsewhere is untested, and it is the number everything here is scaled by.

One quantization scheme. MLX 4-bit group quantization at one bit-width. Duan sweep INT4–INT8; we do not.

One architecture family for the ladder. The floor is cross-architecture but the rungs are all Llama.

Matching is one operationalisation of “same feature”. Reciprocal nearest neighbour on activation fingerprints is not the only reasonable choice, and agreement percentages are not invariant to it—as the Pearson/cosine spread in Table 1 shows within this paper alone.

9 Reproducibility

Code and data: <https://github.com/american-code/ResearchPapers>. Archived at **[PENDING: Zenodo concept DOI]**.

artifact	path
run script	data/sae-runs/run_quant_interp.sh
optimizer arm	data/sae-runs/run_muon_arm.sh
trainer	data/sae-runs/train_sae.py
matching	data/sae-analysis/quant_interp_match.py
seed control	data/seed-stability/

All runs on a single Apple M1 Max, 32 GB unified memory, via MLX. Seed 42 throughout except in the seed control.

References

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